

GIS Notes for Week 1 – Week 5

1. GIS

A Geographic Information System (GIS) is an integrated computer-based system designed to capture, store, manage, manipulate, analyze, model, and visualize geographically referenced data (spatial data) along with their associated attribute information for effective decision-making. In simple terms, GIS links location (where) with information (what) and enables spatial analysis.

Key Characteristics of GIS

- Handles spatial and non-spatial data together
- Uses coordinate systems for referencing
- Performs spatial analysis and modeling
- Produces maps, reports, and thematic outputs

Functions of GIS

- Data input
- Data storage
- Data manipulation
- Spatial analysis
- Visualization

Applications

- Urban planning
- Environmental monitoring
- Transportation engineering
- Disaster management
- Water resource planning

2. Raster Data Compression

Introduction to Raster Data

A raster data model represents spatial data as a matrix of rows and columns (grid cells or pixels) where each cell contains a value representing a geographic attribute such as elevation, temperature, land use, or reflectance. In large GIS and remote sensing applications, raster datasets (satellite images, DEMs, thematic maps) can occupy very large storage space, especially when:

- Resolution is high (small pixel size)
- Coverage area is large
- Multi-band imagery is used
- Hence, raster data compression becomes essential.

Definition of Raster Data Compression

Raster data compression is the process of reducing the storage size of raster datasets by eliminating redundancy in pixel values while preserving necessary spatial and attribute information.

It improves:

1. Storage efficiency
2. Data transmission speed
3. Processing performance

Why Raster Compression is Required?

Raster data consumes large storage because:

- Each pixel stores a value (8-bit, 16-bit, 32-bit etc.)
- High-resolution satellite images contain millions of pixels
- Multi-spectral images have multiple bands

For example:

If: Image size = $10,000 \times 10,000$ pixels

Each pixel = 1 byte

Storage required: $10,000 \times 10,000 = 100,000,000$ bytes = 100 MB

For multi-band imagery, storage multiplies.

Thus, compression is necessary.

Types of Raster Data Compression

Raster compression is classified into two main categories:

1. Lossless Compression

Lossless compression reduces file size without any loss of data. The original dataset can be reconstructed exactly.

Characteristics:

- No information loss
- Suitable for scientific analysis
- Used in GIS modeling and DEMs

Common Techniques:

(1) Run-Length Encoding (RLE)

Stores consecutive repeated pixel values as:

Value + Number of repetitions

Example:

Original row: 1 1 1 1 1 2 2 2 3 3

Compressed form: (1,5) (2,3) (3,2)

Best for:

- Land use maps
- Thematic classified maps

(2) LZW (Lempel-Ziv-Welch)

- Dictionary-based compression
- Replaces repeated patterns with shorter codes
- Used in TIFF format

Advantages:

- Efficient
- Widely supported in GIS software

(3) Huffman Coding

- Statistical compression
- Short codes assigned to frequent values
- Longer codes to rare values

(4) DEFLATE Algorithm

- Combination of LZ77 + Huffman coding
- Used in PNG format

2. Lossy Compression

Lossy compression reduces file size by removing some data permanently.

Characteristics:

- Some information is lost
- Not suitable for precise analysis
- Suitable for visualization and web mapping

Common Techniques:

(1) JPEG Compression

- Uses Discrete Cosine Transform (DCT)
- Removes high-frequency components
- Good for aerial photographs

(2) Wavelet Compression (JPEG 2000)

- Advanced mathematical transformation
- Better compression ratio
- Maintains higher image quality

Raster Compression Based on Data Structure:

a) Pixel-Based Compression

Operates on individual pixels.

Examples:

- RLE
- Huffman coding

b) Block-Based Compression

Divides image into blocks and compresses each block separately.

Example:

- JPEG

c) Transform-Based Compression

Transforms data to frequency domain.

Example:

- DCT (JPEG)
- Wavelet transform

Compression Ratio

Compression Ratio (CR) is: $\frac{\text{Original File Size}}{\text{Compressed File Size}}$

Higher ratio = Better compression.

Example:

Original: 100 MB, Compressed: 25 MB

CR = 100 / 25 = 4:1

Factors Affecting Raster Compression

- Data type (continuous vs categorical)
- Spatial resolution

- Number of bands
- Bit depth
- Data redundancy
- Presence of noise
- Applications of Raster Compression
- Satellite image storage
- Web GIS applications
- Digital Elevation Models (DEM)
- Climate data storage
- Noise mapping and pollution mapping
- Remote sensing analysis

Advantages of Raster Compression

- Reduces storage cost
- Improves data transmission
- Faster data loading
- Efficient database management
- Reduces server load in Web GIS

Limitations of Raster Compression

- Lossy compression may reduce accuracy
- Repeated compression causes degradation
- Complex algorithms require processing power
- Not all formats support advanced compression

Role in GIS Analysis

- Compression affects:
- Overlay analysis
- Raster algebra
- Terrain modeling

Hydrological analysis

- Lossless compression must be used for:
- DEM analysis
- Slope calculation
- Hydrological modeling

3. Major GIS Functions and Their Importance

GIS functions refer to the core operations performed by a GIS system to process and analyze spatial data.

Major Functions:

1. Data Input

- Digitizing maps
- GPS survey
- Satellite imagery import

2. Data Storage and Management
 - Database management systems
 - Data organization
3. Data Manipulation
 - Projection transformation
 - Reclassification
 - Editing
4. Spatial Analysis
 - Buffer analysis
 - Overlay analysis
 - Network analysis
 - Interpolation
5. Visualization
 - Thematic maps
 - 3D surface modelling

Importance

- Supports planning decisions
- Enhances infrastructure management
- Assists environmental assessment
- Improves resource allocation

4. Interpolation in GIS

In GIS, spatial data is often collected at discrete sample locations (e.g., bore wells, weather stations, noise monitoring points). However, for analysis and decision-making, we frequently require a continuous surface representation.

Interpolation is the technique used to estimate unknown values at unsampled locations using known data points.

Definition of Interpolation

Spatial interpolation is a GIS analytical method used to predict or estimate values of a variable at unmeasured locations based on values measured at known sample points, assuming spatial continuity.

It converts: Discrete point data → Continuous raster surface

Principle of Interpolation

Interpolation is based on the concept of spatial autocorrelation, which states:

“Things that are close to each other are more similar than those that are farther apart.”

This principle is fundamental in geography and spatial statistics.

Why Interpolation is Required?

In practical scenarios:

- Rainfall measured at few stations → Need rainfall map
- Groundwater depth at bore wells → Need groundwater contour map
- Noise level at 50 locations → Need city-wide noise map
- Air pollution at monitoring stations → Need pollution distribution map

Since it is impossible to measure every location, interpolation estimates unknown areas.

Types of Interpolation Methods

Interpolation methods are broadly classified into:

- A. Deterministic Methods
- B. Geostatistical Methods

A. Deterministic Interpolation Methods

Deterministic methods estimate values using mathematical formulas based on surrounding known points.

1. Inverse Distance Weighting (IDW)

Concept: Nearby points have more influence than distant points.

Characteristics:

- Simple and easy
- Fast computation
- No statistical assumptions

Advantages:

- Easy to understand
- Good for dense datasets

Limitations:

- Produces circular patterns
- Does not account for spatial trend

2. Spline Interpolation

Concept: Fits a smooth mathematical surface passing through known points.

Types:

- Regularized spline
- Tension spline

Suitable For:

- Smooth surfaces like elevation
- Temperature distribution

Advantage:

- Produces smooth surface.

Limitation:

- May overshoot extreme values.

B. Geostatistical Interpolation Methods

These methods consider statistical properties and spatial autocorrelation.

3. Kriging

Definition: Kriging is an advanced geostatistical interpolation technique that estimates unknown values using weighted linear combination of known values based on spatial correlation.

Key Concept: Variogram

A variogram measures spatial dependence:

- Nugget → Random error
- Sill → Total variance
- Range → Distance of influence

Types of Kriging:

- Ordinary Kriging
- Universal Kriging

- Simple Kriging
- Indicator Kriging

Advantages:

- Provides estimation error
- More accurate
- Accounts for spatial structure

Limitations:

- Complex
- Requires statistical expertise
- Computationally intensive

Applications of Interpolation

- Hydrology (water table mapping)
- Environmental pollution mapping
- Climate studies
- Agriculture planning
- Urban heat island studies
- Terrain modelling

Advantages of Interpolation

- Converts point data into continuous surface
- Helps in prediction
- Supports decision-making
- Identifies spatial patterns
- Reduces field survey cost

Limitations of Interpolation

- Accuracy depends on sample density
- Sensitive to outliers
- Method selection affects result
- Assumes spatial continuity
- Edge effects near boundary

Factors Affecting Interpolation Accuracy

- Number of sample points
- Distribution of sample points
- Spatial variability
- Choice of interpolation method
- Parameter settings

Validation of Interpolation

- Cross-validation method:
- Remove one point
- Predict its value
- Compare predicted vs actual
- Compute RMSE (Root Mean Square Error)

Lower RMSE = Better model

5. Topology in GIS

In GIS, spatial data not only represents location but also the spatial relationships between geographic features. These relationships are essential for accurate spatial analysis.

Topology is a fundamental concept in vector data models that ensures spatial features are logically connected and correctly structured.

Topology in GIS refers to the mathematical and logical framework that defines and maintains spatial relationships between geographic features such as points, lines, and polygons.

It describes how spatial features are:

- Connected
- Adjacent
- Contained
- Related to one another

Need for Topology

Without topology:

- Roads may not connect properly.
- Polygons may overlap or have gaps.
- Boundaries may be duplicated.
- Network analysis may fail.

Topology ensures:

- Logical consistency
- Accurate analysis
- Data integrity

Topological Relationships: There are three primary topological relationships:

1. Connectivity: Describes how line features connect at nodes.

Example:

- Road networks
- Water pipelines
- Electric supply networks

Importance:

- Used in route analysis
- Essential for network modelling

2. Adjacency (Contiguity): Describes neighboring relationships between polygons.

Example:

- Adjacent land parcels
- Neighboring districts

Importance:

- Used in regional planning
- Important in spatial statistics

3. Containment: Describes when one feature lies within another.

Example:

- Village inside district
- Lake within forest boundary

Importance:

- Used in administrative boundary analysis

- Essential in overlay operations

Basic Topological Elements

Topology in vector data is built using:

- Nodes

Points at the start and end of lines.

- Arcs (Edges)

Line segments connecting nodes.

- Polygons

Closed areas formed by arcs.

- Vertices

Intermediate points defining shape of lines.

Topology in GIS refers to the spatial relationships between geographic features, such as connectivity, adjacency, and containment.

Types of Topological Relationships

- Connectivity – Roads connected at junctions
- Adjacency – Neighbouring land parcels
- Containment – Village within district

Importance

- Ensures spatial accuracy
- Prevents overlaps and gaps
- Supports network analysis
- Maintains data integrity

Topology Rules in GIS

GIS software allows defining topology rules such as:

For Polygons:

- Must not overlap
- Must not have gaps
- Must be covered by boundary

For Lines:

- Must not self-intersect
- Must not have dangles
- Must not overlap

For Points:

- Must be covered by line
- Must be inside polygon

Advantages of Topology

- Ensures spatial accuracy
- Prevents duplication of boundaries
- Supports network analysis
- Enables adjacency analysis
- Maintains data integrity

- Reduces storage redundancy

Applications of Topology

- Road network routing
- Utility management systems
- Cadastral mapping
- Administrative boundary mapping
- Water distribution network analysis
- Sewer system planning

Topology in Network Analysis

Topology enables:

- Shortest path analysis
- Service area analysis
- Flow direction modeling
- Connectivity analysis

Importance of Topology in Spatial Analysis

Topology is crucial for:

- Overlay analysis
- Buffer analysis
- Network routing
- Spatial queries
- Urban planning decisions

Limitations of Topology

- Complex data structure
- Slower editing process
- Requires careful data validation
- More computational overhead

6. Slope, Aspect, Precision, Data Quality, Model

Slope

- Rate of change of elevation over horizontal distance.
- Measured in degrees or percentage.
- Used in drainage and highway design.

Aspect

- Direction of steepest slope (N, S, E, W).
- Important in agriculture and solar radiation studies.

Precision

Degree of exactness in measurement.

High precision means less random error.

Data Quality

Indicates reliability of spatial data based on:

- Accuracy
- Completeness
- Consistency
- Timeliness

Model

- A simplified representation of real-world systems.

Example: Flood simulation model.

7. Key Components of GIS

The essential elements required for the functioning of a GIS system.

Five Components

Hardware – Computers, GPS, servers

Software – ArcGIS, QGIS

Data – Spatial and attribute

People – Analysts, engineers

Methods – Procedures and workflows

All components must work together for efficient GIS implementation.

8. Raster Data Model

Raster data model represents spatial data as a grid of equal-sized cells, each containing a value representing a geographic attribute.

Structure

- Rows and columns
- Each cell = pixel
- Resolution defines cell size

Advantages

- Simple structure
- Suitable for continuous data
- Easy overlay operations

Limitations

- Large storage
- Low boundary precision

9. TIN Data Model

TIN (Triangulated Irregular Network) is a vector-based surface model that represents terrain using connected non-overlapping triangles formed from irregularly spaced points.

Advantages

- Accurate terrain modeling
- Efficient for variable surfaces
- Stores breaklines

Applications

- Slope analysis
- Watershed modeling
- 3D terrain visualization

10. Data Quality Issues

Types

- Positional accuracy
- Attribute accuracy
- Temporal accuracy
- Logical consistency

- Completeness

Poor data quality leads to:

- Wrong planning decisions
- Incorrect site selection
- Financial losses

Spatial vs Non-Spatial Data

Spatial Data

- Contains geographic coordinates.
Example: Location of bore wells.

Non-Spatial Data (Attribute Data)

- Descriptive information linked to spatial features.
Example: Depth of bore well.

Pre-processing of Spatial Data

Pre-processing refers to cleaning, correcting, and transforming spatial data before analysis.

Steps

- Georeferencing
- Projection transformation
- Digitization
- Topology correction
- Removing duplicates

Importance

Ensures reliable and accurate analysis results.